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AIM: Data Visualization with Matplotlib: Data visualization on a dataset using matplotlib and seaborn libraries

- Scatter plot
- Line plot
- Bar plot
- Histogram
- Box plot
- Pair plot

✓ Theory

1.What is the importance of data visualization in data analysis? How does visualization help in exploratory data analysis (EDA)?

Data visualization is one of the most important parts of data analysis because it helps convert raw numerical data into visual forms such as graphs, charts, and plots. Visual representation makes the data easier to understand and interpret.

In Exploratory Data Analysis (EDA), visualization helps to:

*identify trends and patterns *detect outliers and anomalies *understand data distribution *compare different variables *observe relationships between features *support decision-making

For example, a scatter plot can show correlation between two variables, while a histogram helps in understanding frequency distribution.

2.What is the difference between Matplotlib and other visualization libraries?

Matplotlib is a foundational Python library used for creating static, animated, and interactive visualizations. Other libraries such as Seaborn, Plotly, and Pandas plotting functions are built on top of Matplotlib or provide specialized capabilities.

Difference:

*Matplotlib: Basic and highly customizable plotting library *Seaborn: More attractive statistical graphics, built on Matplotlib *Plotly: Interactive and web-based visualizations *Pandas: Plotting Quick plotting directly from DataFrames

Matplotlib provides more control, while Seaborn makes complex statistical plots easier and more visually appealing.

3.What is the purpose of the pyplot module in Matplotlib?

The pyplot module is a collection of functions in Matplotlib that makes plotting simple and similar to MATLAB commands.

It is commonly imported as:

```
import matplotlib.pyplot as plt
```

The purpose of pyplot is to:

*create figures and axes *plot graphs *label axes *set titles *display plots *customize appearance

Common functions include:

-plt.plot() -plt.scatter() -plt.bar() -plt.hist() -plt.show()

4.Explain the process of creating a basic line plot

The steps to create a basic line plot are:

Import Matplotlib Define x-axis and y-axis data Use plt.plot() Add labels and title Display graph using plt.show()

Example:

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5] y = [10, 20, 25, 30, 40]
```

```
plt.plot(x, y) plt.xlabel("X-axis") plt.ylabel("Y-axis") plt.title("Basic Line Plot") plt.show()
```

5.Why is customization important in data visualization? What customization options are available in Matplotlib?

Customization is important because it improves readability, clarity, and presentation quality of the graph.

It helps in making graphs more informative and visually appealing.

Some customization options in Matplotlib include:

title axis labels line color line style marker style grid legend figure size font size
background style

Example:

```
plt.plot(x, y, color='blue', linestyle='--', marker='o')
plt.title("Customized Plot")
plt.grid(True)
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/train (1).csv")

print(df.head())
```

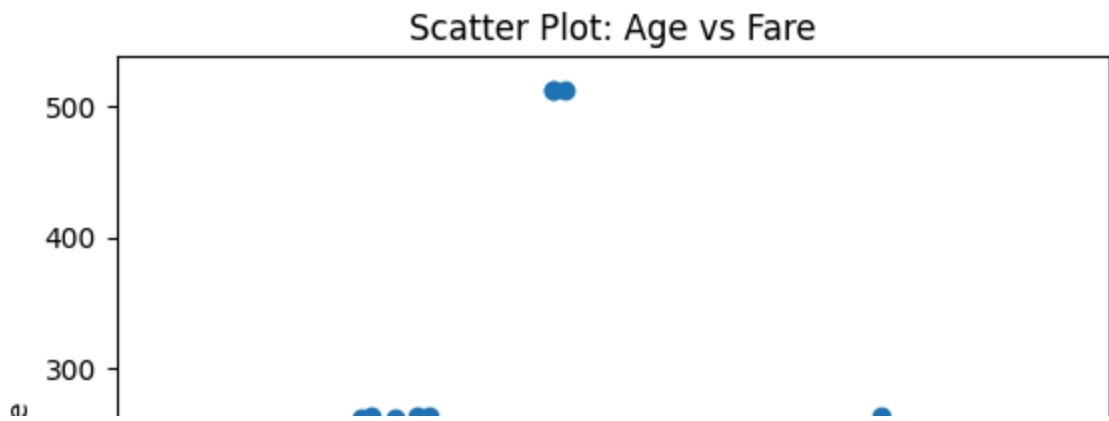
	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp
0	Braund, Mr. Owen Harris	male	22.0	1
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	Heikkinen, Miss. Laina	female	26.0	0
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	Allen, Mr. William Henry	male	35.0	0

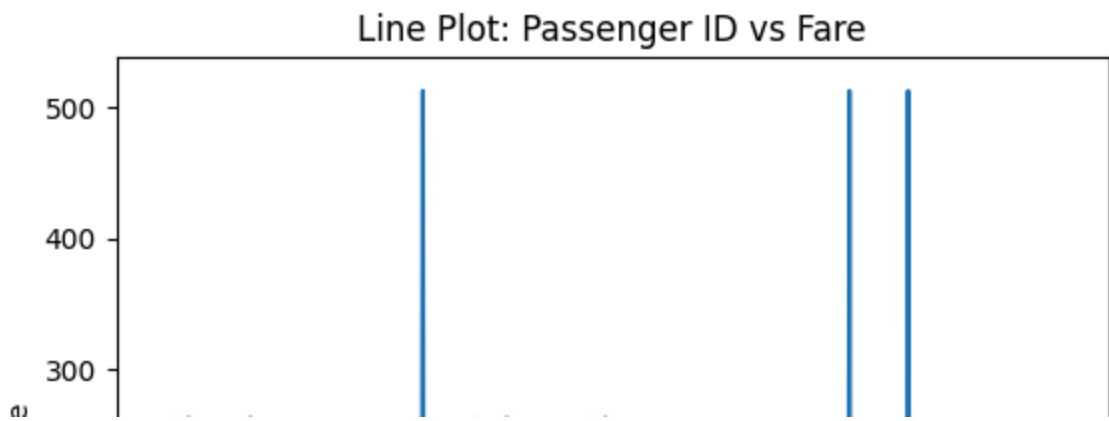
	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
import matplotlib.pyplot as plt

plt.scatter(df['Age'], df['Fare'])
plt.title("Scatter Plot: Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

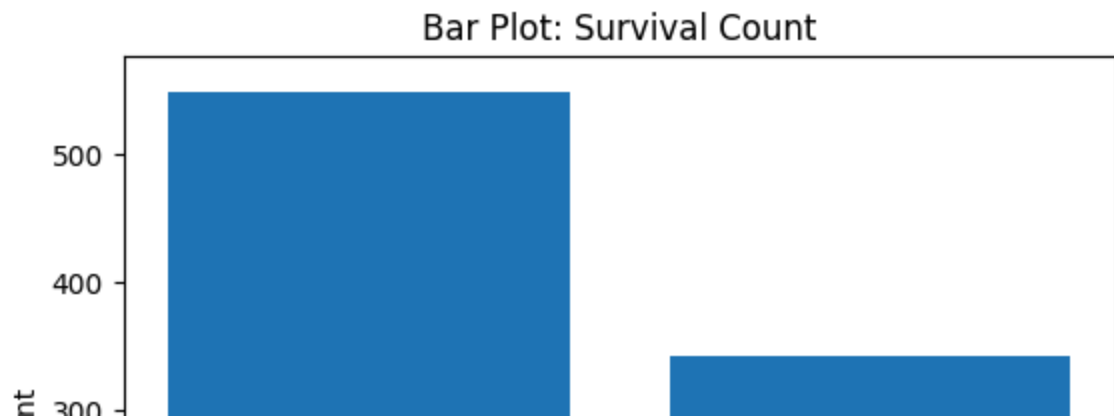


```
plt.plot(df['PassengerId'], df['Fare'])  
plt.title("Line Plot: Passenger ID vs Fare")  
plt.xlabel("Passenger ID")  
plt.ylabel("Fare")  
plt.show()
```



```
survival_count = df['Survived'].value_counts()

plt.bar(survival_count.index, survival_count.values)
plt.title("Bar Plot: Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```



```
plt.boxplot(df['Fare'])
plt.title("Box Plot: Fare Distribution")
plt.ylabel("Fare")
plt.show()
```

Box Plot: Fare Distribution



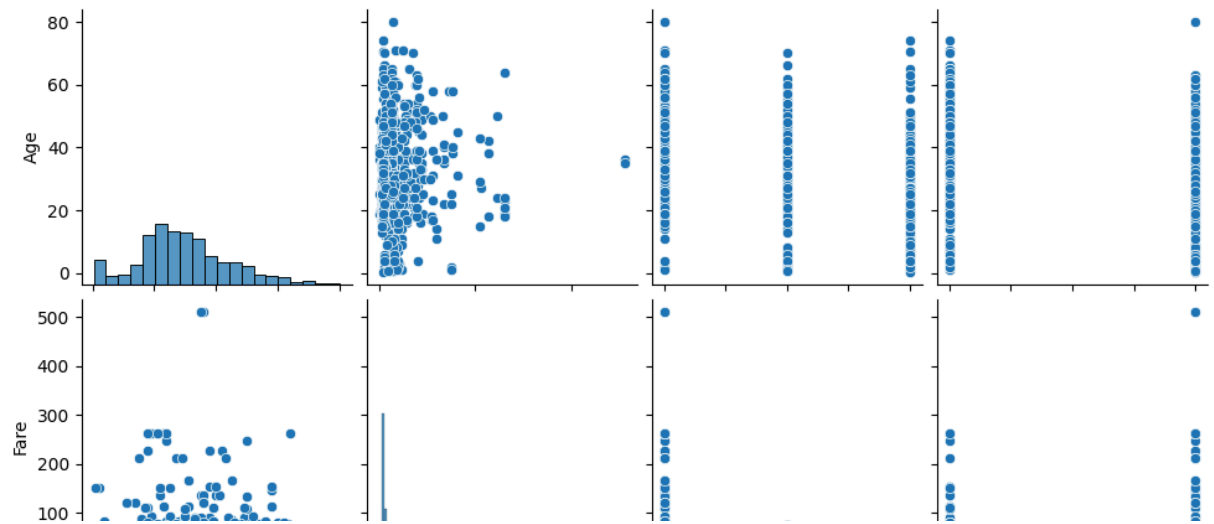
```
plt.hist(df['Age'].dropna(), bins=10)
plt.title("Histogram: Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

Histogram: Age Distribution



```
import seaborn as sns
```

```
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']].dropna())  
plt.show()
```



Conclusion:

In this assignment, different types of data visualization techniques were performed using Matplotlib and Seaborn. Various plots such as scatter plot, line plot, bar plot, histogram, box plot, and pair plot were created successfully. These visualizations help in understanding data patterns, trends, relationships, and distributions more effectively, making data analysis and EDA easier and more meaningful.